The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
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11 Substrates and PCBs
12 Test and Assembly
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14 **Data Centers and Power**

15 The Economics of AI Chips
16 Who Controls the Chain

Chapter 14 Stage 14 of 14 High concentration · Moderately hard to substitute

Data Centers and Power

SemiAnalysis forecast in October 2026 that 50 GW of data center capacity would be added in 2027, about as much as the world had built through 2024. The large transformers that connect such sites to the grid commonly took three years to arrive in 2024.

Specimen 14 · Data Centers and Power

A liquid-cooled rack of 72 GPUs, Nvidia's GB200 NVL72, drawn to scale. It is 2.24 m tall and 600 mm wide, taller than a door. It weighs about 1.36 metric tons, more than a small car, and draws about 120 kW. The newer GB300 NVL72 draws up to 142 kW. The gray steel power shelves and the copper busbar they feed are picked out.

In plain terms

A data center feeds electricity to the machines inside it and carries away the heat they give off. One rack of AI chips, a cabinet taller than a door, can draw 142 kilowatts[1]. Every watt of it comes back out as heat, and liquid piped through the rack carries the heat away[1]. A large training site already holds more than a hundred thousand chips[2]. The newest sites are planned in gigawatts[3]. One gigawatt is the whole output of a large power station, and Epoch AI projects that OpenAI's seven Stargate sites in the United States will add up to more than nine of them by 2029[3]. Getting that much electricity to the site is harder than putting up the building. Transformers and heavy switches connect a site to the grid, and turbines make the power. All of them come from a few suppliers and take years to arrive.

In short

Measured by capacity, US factories supplied 29.2 percent of the large power transformers bought in the US in 2023, and imports the rest[4]. GE Vernova's gas equipment backlog and reserved factory slots stood at 116 GW in mid-2026, against planned output of 20 GW a year from the third quarter[5].

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com
2 Colossus 1, AI data centers directory. Epoch AI, accessed 9 September 2026. epoch.ai
3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai
4 Large Power Transformers from South Korea, Investigation No. 731-TA-1189... U.S. International Trade Commission, August 2024, p. 90. usitc.gov
5 GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission, 22 July 2026. sec.gov

ERCOT, the grid operator for most of Texas, counted 9,456 MW of big power users approved to energize in its overview for September 2026[6]. Judging by peak monthly use in September, it believed 4,132 MW of that was operating[6]. On 3 August 2026 Governor Greg Abbott ordered an audit of every data center seeking a connection before any data center project moves forward[7]. In September 2026 ERCOT said it had paused approvals to energize data centers of 75 MW or more until the audit and a review of their impact on local communities are complete[8].

A wafer passes through a leading-edge fab in months, but in 2024 a large power transformer commonly took three years to arrive[9]. Before the pandemic, the same order took under a year[10].

Chokepoint card Data Centers and Power, stage 14 of 14

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Nvidia named nine data center power system providers in October 2025 for its 800-volt designs for gigawatt-scale data centers[11].

Price or market size

$580 billion

of data center investment expected worldwide in 2025, in the International Energy Agency's November 2025 outlook[12]

Who leads	US	**GE Vernova**	Gas equipment, transformers and switchgear, including through Prolec GE[13]
	CH	**Hitachi Energy**	Power transformers and switchgear[13]
	DE	**Siemens Energy**	Grid technologies, transformers, turbines[13]
	JP	**Mitsubishi Power**	Power generation equipment, including gas turbines[13]
	US	**Vertiv**	Data center power systems[11]
	FR	**Schneider Electric**	Cooling, through its Motivair division[14]
Where it is made	US	**United States**	The largest data center operators, and GE Vernova turbines and grid equipment
	CH	**Switzerland**	Hitachi Energy transformers and switchgear, and ABB switchgear
	DE	**Germany**	Siemens Energy grid technologies and turbines
	JP	**Japan**	Mitsubishi Power equipment, including gas turbines
Why substitution is possible	Buyers of grid equipment can choose among suppliers, and GE Vernova's 2025 annual report names Hitachi Energy, Siemens Energy, Schneider Electric and ABB among its key competitors[13]. But in the United States in 2024 the longest large power transformer lead times reached 60 months[9]. An International Energy Agency survey published in 2025 found waits of up to four years worldwide[15].		

6 Electric Reliability Council of Texas, ERCOT Monthly Operational Overview... Electric Reliability Council of Texas, 16 October 2026, p. 9. ercot.com

7 Governor Abbott Directs Comprehensive Data Center Audit. Office of the Texas Governor, 3 August 2026. gov.texas.gov

8 Electric Reliability Council of Texas, Item 14: Batch Zero Update, Board of... Electric Reliability Council of Texas, 11 September 2026. ercot.com

9 U.S. Department of Energy Large Power Transformer Resilience Report to... U.S. Department of Energy, 10 July 2024, p. 8. energy.gov

10 U.S. Department of Energy Large Power Transformer Resilience Report to... U.S. Department of Energy, 10 July 2024, p. 18. energy.gov

11 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

12 Executive summary – World Energy Outlook 2025. International Energy Agency, 12 November 2025. iea.org

13 GE Vernova Inc., Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 29 January 2026. sec.gov

14 Hotter Than a Hot Tub: The 45°C Breakthrough to Cool AI's Biggest Machines. Nvidia, 22 June 2026. blogs.nvidia.com

15 Executive summary – Building the Future Transmission Grid. International Energy Agency, 2025. iea.org

Where China stands

China used 25 percent of the world's data center electricity in 2024, second to the United States, by the International Energy Agency's estimate [16]. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024 [17].

Where the US stands

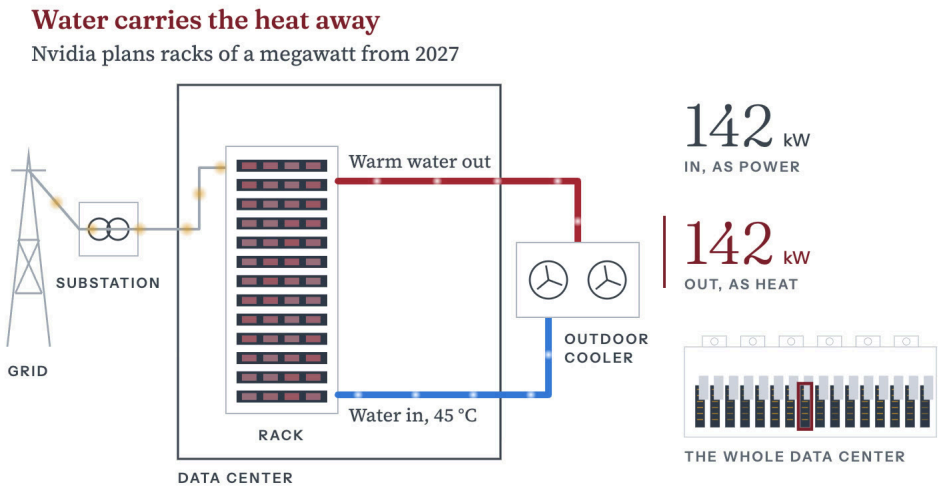
US factories supplied 29.2 percent of the large power transformers bought in the United States in 2023, measured by capacity [4]. Constellation said in August 2026 that it expects to restart the Crane nuclear plant in 2027 [18]. Microsoft has agreed to buy the plant's power [19].

01

How it works

Figure 14.1 · Explainer, 21 seconds

Power in, heat out



1. **The site.** Data centers are where AI chips go to work. A large training site holds a hundred thousand of them, in rows of racks.
2. **Power in.** Power comes in from the grid, stepped down by a substation, and runs through the data center to every rack. One rack of AI chips can draw 142 kilowatts.
3. **Heat.** Every watt of it comes back out as heat, more than air can carry away.
4. **Water out.** So water runs through plates on the chips, carries the heat to coolers outdoors and comes back. Nvidia plans for racks of a megawatt each from 2027, so the power coming in and the heat going out keep growing.

Note: Simplified diagram, with illustrative sizes. An Nvidia GB300 NVL72 rack draws at most 142 kW. Nvidia gives 45 °C as the supply-water temperature for its Vera Rubin racks. The megawatt racks from 2027 are Nvidia's plan.

The rack is the basic unit of AI compute. Nvidia's GB300 NVL72 rack draws up to 142 kW through eight 33 kW power shelves, the rack's own power supplies [1]. Large sites hold

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com

4 Large Power Transformers from South Korea, Investigation No. 731-TA-1189... U.S. International Trade Commission, August 2024, p. 90. usitc.gov

16 Executive summary – Energy and AI. International Energy Agency, 10 April 2025. iea.org

17 Executive summary – Electricity 2025. International Energy Agency, 14 February 2025. iea.org

18 Constellation Reports Second Quarter 2026 Results. Constellation Energy, 6 August 2026. constellationenergy.com

19 Constellation Ahead of Schedule for Launch of Crane Clean Energy Center. Constellation Energy, 19 February 2025. constellationenergy.com

many such racks. Colossus 1, an AI data center in Memphis, Tennessee, held an estimated 230,000 Nvidia GPUs by July 2025[2].

Power reaches the rack through a chain of heavy equipment, and each piece has its own lead time, the wait between order and delivery. Between the grid and the rack, a substation steps the high voltage down with transformers. Switchgear, the building's heavy circuit breakers, can cut any part of the site off in a fault. Battery-backed supplies keep the racks running for the seconds it takes backup generators to start. Distribution boards split the feed among the rows of racks.

At the back of each rack, a busbar carries the current down to the server trays. It is a solid copper bar, because an ordinary cable would overheat at that current.

Heat leaves in water. Cold plates, metal blocks with coolant running through them, sit on the GPU packages and pass the heat to the building's water loop, which releases it outdoors[14]. Nvidia's Rubin racks run coolant at up to 45 °C, warm enough that in many climates the loop can release its heat without turning on chillers, the refrigeration machines that cool water[14].

Nvidia is also moving the power that feeds its racks to 800 volts of direct current. Raising the voltage cuts the current needed for the same power, and the heat in copper comes from current. Nvidia said in October 2025 that at 800 volts the same wire can carry 157 percent more power than at 415 volts of alternating current[20]. Inside racks such as the GB300 NVL72, power runs at 54 volts, and at that voltage a one-megawatt rack would need up to 200 kg of copper busbar[21].

02

Variants and trade-offs

Cooling

- **Air.** Heat exchangers in the rack door work when density, the power each rack draws, is moderate, but Nvidia's GB300 NVL72 rack is liquid cooled[1].
- **Direct-to-chip liquid.** Nvidia's Rubin platform is 100 percent liquid cooled, and Nvidia says every cloud provider and data center operator building for it is moving to liquid[14].
- **Immersion.** The servers sit in a bath of fluid, which makes them hard to service.

Nvidia says the cooling supply chain is keeping pace with the move to liquid[14]. Motivair, Schneider Electric's cooling division, has worked with Nvidia for nearly a decade, following Nvidia's product roadmap[14].

Rack power

Nvidia's 800-volt DC architecture is designed for racks from 100 kW to more than a megawatt, with megawatt racks from 2027[21]. Nvidia is making the change because the

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com
2 Colossus 1, AI data centers directory. Epoch AI, accessed 9 September 2026. epoch.ai
14 Hotter Than a Hot Tub: The 45°C Breakthrough to Cool AI's Biggest Machines. Nvidia, 22 June 2026. blogs.nvidia.com
20 Building the 800 VDC Ecosystem for Efficient, Scalable AI Factories. Nvidia, 13 October 2025. developer.nvidia.com
21 NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories. Nvidia, 20 May 2025. developer.nvidia.com

54-volt distribution in existing racks begins to hit physical limits once racks exceed 200 kW[21].

Nvidia names nine data center power system providers for its 800-volt designs for gigawatt-scale data centers, among them ABB, Eaton, Hitachi Energy, Mitsubishi Electric, Schneider Electric, Siemens and Vertiv[11].

A megawatt rack changes the busbar, the floor loading, the coolant flow, the leak detection and the energy stored in the rack. Nvidia's Vera Rubin NVL72 rack design adds energy storage to keep power steady[11].

Where the electrons come from

- **Grid interconnection.** The slow path in the United States, where the queues to connect are long[3].
- **Behind-the-meter gas**, generation on the site itself. The fast path, which at least three of OpenAI's seven US Stargate sites will use to avoid grid queues[3].
- **Nuclear.** Existing plants already sell power to data centers[22]. Talen Energy supplies Amazon's data center campus next to the Susquehanna nuclear plant in Pennsylvania under a June 2025 contract expected to reach 1,920 MW no later than 2032[22]. Restarts and new reactors are still to come. Constellation signed a power purchase agreement with Microsoft to restart Three Mile Island Unit 1 as the 835 MW Crane Clean Energy Center[19]. The plant shut in 2019[23]. A $1 billion federal loan that closed in November 2025 partly finances its restart, which is subject to licensing approvals from the US Nuclear Regulatory Commission[23].

03

Who makes it

The largest data center operators pay for the buildout. Capital spending by Alphabet, Amazon, Meta, Microsoft and Oracle came to $448.2 billion in 2025, the sum of quarterly figures compiled by Epoch AI, a research group[24]. If the trend continues, it will reach $770 billion in 2026[24]. The International Energy Agency expected worldwide investment in data centers to reach $580 billion in 2025[12].

3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai

11 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

12 Executive summary – World Energy Outlook 2025. International Energy Agency, 12 November 2025. iea.org

19 Constellation Ahead of Schedule for Launch of Crane Clean Energy Center. Constellation Energy, 19 February 2025. constellationenergy.com

21 NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories. Nvidia, 20 May 2025. developer.nvidia.com

22 Talen Energy Corp, Form 8-K current report for the period ended 2025-06-11 (8-K). U.S. Securities and Exchange Commission, 11 June 2025. sec.gov

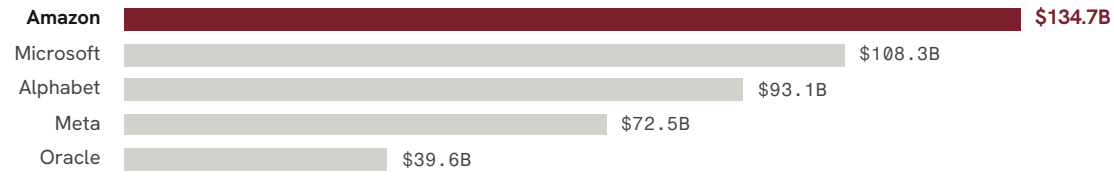
23 Energy Department Closes Loan to Restart Nuclear Power Plant in Pennsylvania. U.S. Department of Energy, 17 November 2025. energy.gov

24 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI, 26 February 2026. epoch.ai

Figure 14.2

Amazon spent the most of five large operators, $134.7 billion in 2025

Capital spending by five large US data center operators, 2025



Source: Epoch AI, hyperscaler capex trend, February 2026, quarterly figures from company filings summed for 2025 [24]

Epoch AI also projects that OpenAI's seven US Stargate sites will exceed 9 GW by 2029[3]. It expects the site in Abilene, Texas, to grow from an estimated 0.3 GW in April 2026 to 1.2 GW in the fourth quarter of 2026[3].

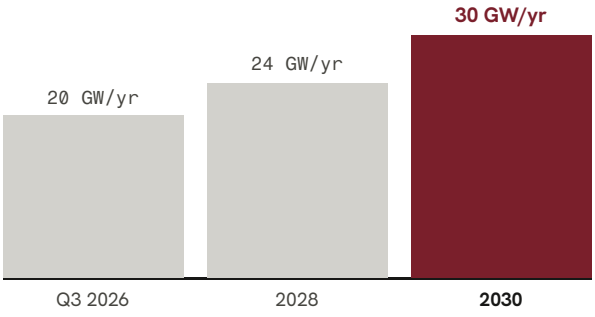
A few large firms make the turbines, transformers and switchgear. GE Vernova builds large gas turbines, and its 2025 annual report names Siemens Energy and Mitsubishi Power among the key competitors of its power generation business[13]. Its electrification business makes power transformers and switchgear, and the report names Hitachi Energy, Siemens Energy, Schneider Electric and ABB among that business's key competitors[13].

Equipment makers limit how fast the buildout can go. GE Vernova's gas equipment backlog and reserved factory slots grew from 100 GW to 116 GW in the second quarter of 2026[5]. In the first half of 2026, data center orders for its electrification business, which makes grid equipment, passed $5 billion, more than double that business's data center orders for all of 2025[5]. GE Vernova bought the remaining half of Prolec GE, a transformer maker, in the first half of 2026[5].

Figure 14.3

GE Vernova plans 30 GW of annual gas turbine output

GE Vernova's planned annual gas turbine output



Source: GE Vernova second-quarter 2026 results, July 2026 [5]

3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai

5 GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission, 22 July 2026. sec.gov

13 GE Vernova Inc., Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 29 January 2026. sec.gov

24 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI, 26 February 2026. epoch.ai

04

Capacity and demand

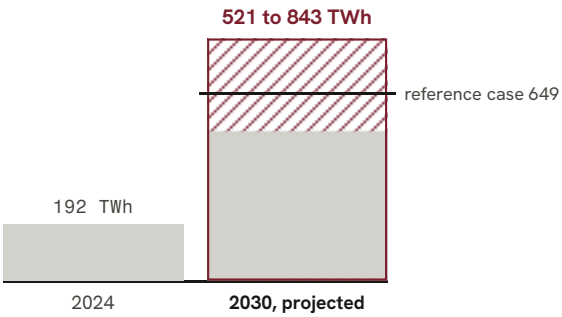
US data center electricity use is projected to rise steeply by 2030. US data centers used 192 TWh in 2024, 4.7 percent of national electricity, according to a June 2026 estimate by Lawrence Berkeley National Laboratory that leaves out cryptocurrency mining[25]. The lab's reference case, its main scenario, projects 464 TWh in 2028 and 649 TWh in 2030, which would be 11.8 percent of forecast US electricity use[25]. Its scenarios for 2030 range from 521 to 843 TWh, or 9.5 percent to 15.3 percent[26].

The US Energy Information Administration expects electricity demand to keep rising, but in September 2026 it cut its 2027 forecast for the region that includes Texas. In its October 2026 Short-Term Energy Outlook, it expects US electricity sales of 4,152.0 TWh in 2026 and 4,216.6 TWh in 2027[27]. It expects electricity demand from the commercial sector, which includes data centers, to grow 2.8 percent in 2027 and demand from industry to grow 2.7 percent[28]. Its September outlook had lowered the 2027 electricity sales forecast for the region that includes Texas by 4.7 percent from August[29]. That outlook noted that Texas had paused connecting new data centers to the grid while the state audits proposed projects[30]. The pause, set by the grid operator for most of Texas, covers data centers and crypto mining sites of 75 MW or more[8].

Figure 14.4

US data centers' projected 2030 use ranges from 521 to 843 TWh

US data center electricity use, 2024 and 2030 projection range



Source: Lawrence Berkeley National Laboratory, June 2026 [26]

By the International Energy Agency's estimate, China accounted for 25 percent of the world's data center electricity use in 2024, second to the United States at 45 percent[16]. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024, the agency found[17].

8 Electric Reliability Council of Texas, Item 14: Batch Zero Update, Board of... Electric Reliability Council of Texas, 11 September 2026. ercot.com

16 Executive summary – Energy and AI. International Energy Agency, 10 April 2025. iea.org

17 Executive summary – Electricity 2025. International Energy Agency, 14 February 2025. iea.org

25 United States Data Center Energy Usage Report: 2025 Update. Lawrence Berkeley National Laboratory, 18 June 2026, p. 10. energy.gov

26 United States Data Center Energy Usage Report: 2025 Update. Lawrence Berkeley National Laboratory, 18 June 2026, p. 11. energy.gov

27 Short-Term Energy Outlook, October 2026. U.S. Energy Information Administration, 6 October 2026, p. 46. eia.gov

28 Short-Term Energy Outlook, October 2026. U.S. Energy Information Administration, 6 October 2026, p. 12. eia.gov

29 Short-Term Energy Outlook, September 2026. U.S. Energy Information Administration, 8 September 2026, p. 4. eia.gov

30 Short-Term Energy Outlook, September 2026. U.S. Energy Information Administration, 8 September 2026, p. 10. eia.gov

SemiAnalysis, a research firm, forecast in October 2026 that 50 GW of data center capacity would be added in 2027, which it said was about as much as the world's data center market had built in total through 2024[31]. Just 40 campuses in an advanced stage of construction held more than 24 GW, most of it to be delivered in 2027, it found[31].

Better chips do not by themselves cut total energy use. Google says its Ironwood TPU, an AI chip, delivers twice the performance per watt of Trillium, the TPU before it[32]. A token is the chunk of text a model reads or writes. Energy per token can fall while total energy use still rises, if the number of tokens rises faster.

05

The chokepoint

On the Texas grid, big power users have asked for far more electricity than is connected and running. ERCOT, the grid operator for most of Texas, was tracking about 474 GW of large loads (big power users) seeking a connection in June 2026, roughly 90 percent of it data centers[33]. In its operational overview for September 2026, ERCOT counted 9,456 MW of large loads approved to energize, that is, cleared to switch on[6]. Judging by peak monthly use in September, it believed only 4,132 MW of it was operating[6].

Before a big user can connect, engineers must study whether the nearby grid can carry its load. In a preliminary review in July 2026, ERCOT found that about 205 GW of the requests already had such a study, which made them eligible for Batch Zero, the first batch of requests it studies together[34]. In September 2026 ERCOT conditionally classed 66.4 GW of Batch Zero projects as base load and 127.9 GW as studied load, pending final decisions in December[8]. Under ERCOT's Batch Zero rules, base load is demand from projects already running or far enough along to be counted without further study[35]. Studied load is demand that ERCOT will study before giving each project a share of the grid's capacity[35].

Texas has paused new connections for large data centers while it audits them. On 3 August 2026 Texas Governor Greg Abbott ordered the state's Public Utility Commission and ERCOT to audit every data center waiting to connect to the grid before any data center project moves forward[7]. ERCOT then paused approvals to energize data centers and crypto mining sites of 75 MW or more until the audit and a review of their impact on local communities are complete[8]. It will report on the audit and the community review to the Public Utility Commission by 10 December 2026[8].

Projects in that queue also wait on equipment.

6 Electric Reliability Council of Texas, ERCOT Monthly Operational Overview... Electric Reliability Council of Texas, 16 October 2026, p. 9. ercot.com

7 Governor Abbott Directs Comprehensive Data Center Audit. Office of the Texas Governor, 3 August 2026. gov.texas.gov

8 Electric Reliability Council of Texas, Item 14: Batch Zero Update, Board of... Electric Reliability Council of Texas, 11 September 2026. ercot.com

31 Datacenters 2027: The Year of Acceleration, +50GW in a Single Year. SemiAnalysis, 19 October 2026. newsletter.semianalysis.com

32 Inside the Ironwood TPU codesigned AI stack. Google Cloud, 6 November 2025. cloud.google.com

33 Electric Reliability Council of Texas, ERCOT update to the Senate Committee on... Electric Reliability Council of Texas, 29 July 2026, p. 6. ercot.com

34 Electric Reliability Council of Texas, ERCOT update to the Senate Committee on... Electric Reliability Council of Texas, 29 July 2026, p. 10. ercot.com

35 ercot.com

- **Large power transformers.** In the United States in 2024, lead times of 36 months were commonly quoted and the longest reached 60 months, against under a year before the pandemic[9]. By a US Commerce Department estimate, 82 percent of the units put into US service in 2019 were imported[36]. In an industry survey published in 2025, the International Energy Agency found that large power transformers took up to four years to secure worldwide[15]. It also found that their average lead times had almost doubled since 2021[15]. A transformer's size is given as a power rating in megavolt-amperes (MVA)[9]. Measured by capacity, US factories supplied 29.2 percent of the transformers rated 60 MVA and above bought in the US in 2023, and imports supplied the rest[4].
- **Gas turbines.** GE Vernova's gas equipment backlog and reserved factory slots stood at 116 GW in mid-2026[5]. In July 2026 it expected to build gas turbines at a yearly rate of 20 GW in the third quarter of 2026, 24 GW in 2028 and 30 GW in 2030[5].
- **Firm generation**, the kind that runs whatever the weather. The Nuclear Regulatory Commission has approved a fuel license amendment for the Crane plant[18]. Constellation called the approval a major milestone toward restarting the plant in 2027[18].

Memory chips are also in short supply. Micron's quarterly report for the period ended 28 May 2026 says AI-driven growth in demand for memory and storage is outpacing industry supply[37]. The memory shortage and the wait for power equipment both limit how much of the 2026 capital spending turns into working compute.

Four questions to answer from memory

1. Why are AI racks cooled with liquid? `see Cooling`
2. How big are the newest AI data centers? `see How it works`
3. What slows down building an AI data center besides chips? `see The chokepoint`
4. Which two countries used the most data center electricity in 2024? `see Capacity and demand`

Four things to remember

— One AI rack can draw 142 kilowatts, too much heat for air, so racks are cooled with liquid.

— The newest AI sites are planned in gigawatts, and each gigawatt is the output of a large power station.

— Power equipment is one of the slow parts of building a data center. In 2024 lead times of 36 months were commonly quoted for a large transformer.

4 Large Power Transformers from South Korea, Investigation No. 731-TA-1189... U.S. International Trade Commission, August 2024, p. 90. `usitc.gov`

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36 U.S. Department of Energy Large Power Transformer Resilience Report to... U.S. Department of Energy, 10 July 2024, p. 20. `energy.gov`

37 MICRON TECHNOLOGY INC, Form 10-Q quarterly report for the period ended... U.S. Securities and Exchange Commission, 25 June 2026. `sec.gov`

— The United States used 45 percent of the world's data center electricity in 2024 and China 25 percent, by the International Energy Agency's estimate.

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Made by **Yashvardhan Sharma**. Corrections and questions: contact@chipsupplychain.org.

A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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